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MZ 505
Inventory Systems

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Session 08: Newsvendor Model - Applications in SC Contracting & Revenue Management

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Outline

- The **newsvendor model** determines a firm's optimal (i.e., profit maximizing) order quantity when:
  - Demand is random.
  - Product has a single selling season.
  - Single opportunity to procure at the start of the season.
- However, as firms operate within supply chains, a firm's optimal ordering decision may not be optimal for the entire supply chain.
- There are contracts that can help resolve this supply chain coordination issue.
  - A **contract** provides the parameters within which a firm downstream a supply chain places orders and the firm upstream fulfills them.

The double marginalization problem

- Suboptimal supply chain performance occurs because…
  - Each firm makes decisions based on their own margin, not the supply chain's margin.
  - This is called **double marginalization**.
- Let's take a look at a ski jacket example.

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Sports Obermeyer

- Sports Obermeyer is a US-based manufacturer of high-end skiing apparel (jackets, pants, sweaters, fleeces, etc.).
- SKI PRO is one of its retailers in the US.
- Obermeyer manufactures its products in Asia (after receiving orders from retailers), so the replenishment lead time to the US is long.
- The selling season for ski-ware is short and styles change significantly from year to year.
- As a result, SKI PRO places a single order and receives only one delivery of Obermeyer apparel before each season.

Example: Ski Jacket

- **Obermeyer**
  - Production cost: c = $100 / unit
  - Revenue (wholesale price): $w / unit
- **SKI PRO**
  - Retail price: p = $250 / unit
  - Salvage value: s = $75 / unit



- Season's demand
  - Normal (μ=1000, σ=300)
- Assume goodwill cost (g) is zero for both entities.

Wholesale Price Contract

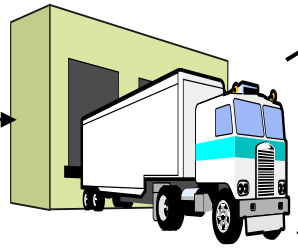
Ski jacket: Wholesale contract

Variable production cost = $100

Wholesale Price =$w









Selling price=$250

Salvage value=$75



Manufacturer

Manufacturer DC

Retail DC

Stores

Adapted from D. Simchi-Levi, 2005

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Ski Jacket: Wholesale contract with w = $175

- Assume that Obermeyer sets w = $175
- SKI PRO orders the optimal (profit maximizing) quantity:
  - $C_u = p - w + g = 250 - 175 = 75$; $C_o = w - s = 175 - 75 = 100$
  - Critical ratio = 75 / 175 = 0.4286
  - $Q^* = F^{-1}(0.4286) = 946$ units
- SKI PRO's expected profit:
  - $\pi_R(Q; w) = pE[Sold] + sE[Left] - gE[Short] - cQ = \$54{,}392$
- Obermeyer's profit:
  - $\pi_S(w; Q(w)) = (w - c)Q = \$70{,}950$
- Supply chain expected profit:
  - $\pi_{SC}(w, Q) = \pi_R + \pi_S = \$125{,}342$

Can we do better?

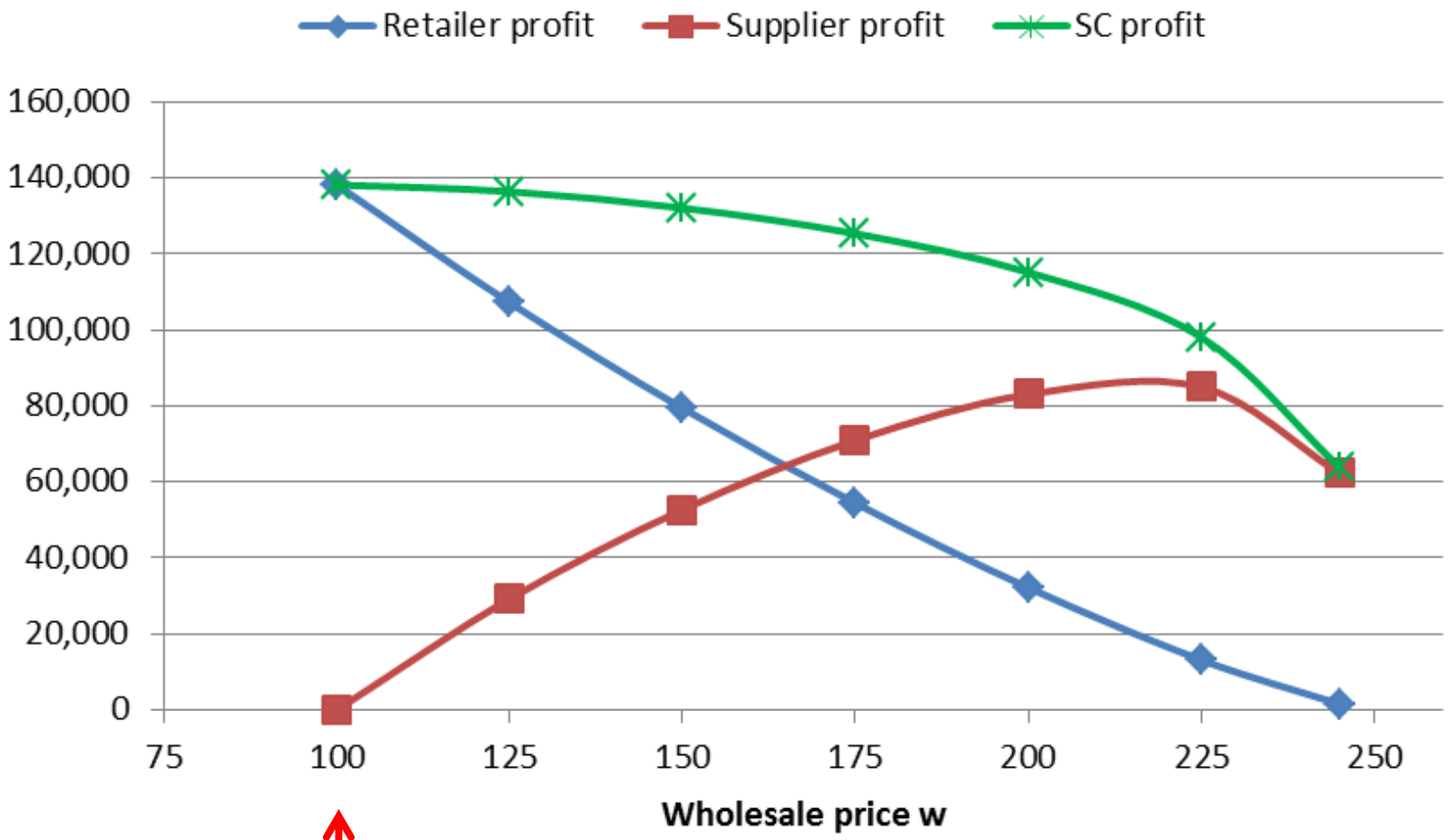
Ski Jacket: Can we do better?

- Consider the integrated supply chain (also called "coordinated SC")
  - I.e., the two firms operating as one entity or having a single decision maker.
- Optimal (profit maximizing) quantity for the integrated firm:
  - $C_u = p - c + g = 250 - 100 = 150$; $C_o = c - s = 100 - 75 = 25$
  - Critical ratio = 150 / 175 = 0.8571
  - $Q^* = F^{-1}(0.8571) = 1{,}320$ units (39.5% more…)
- Integrated firm's expected profit:
  - $\pi_{SC}(Q; w) = pE[Sold] + sE[Left] - gE[Short] - cQ$

    $= \$138{,}154$

… a 10.22% increase

Ski Jacket: The cost of double marginalization



SC optimal (but not feasible)

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Buy-back contracts

A solution to double marginalization: Share risk

- Suppose Obermeyer offers to buy back unsold jackets at $b per unit.
  - SKI PRO incurs a cost (r) to ship the jackets back: r = $5
- Necessary conditions for the buy-back price b to be feasible:
  - b < w (=$175) so that SKI PRO doesn't make profit returning merchandise.
  - b > s + r (=$75 +$5) so that SKI PRO prefers to return rather than salvage.

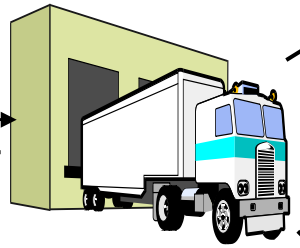
Ski jacket: Buyback contract

Variable production cost = $100

Wholesale Price =$w









Selling price=$250

Salvage value=$75

Return cost=$5



Manufacturer **Manufacturer DC** **Retail DC** **Stores**

Buyback Price =$b

Adapted from D. Simchi-Levi, 2005

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Ski Jacket: Buyback contract with w = $175

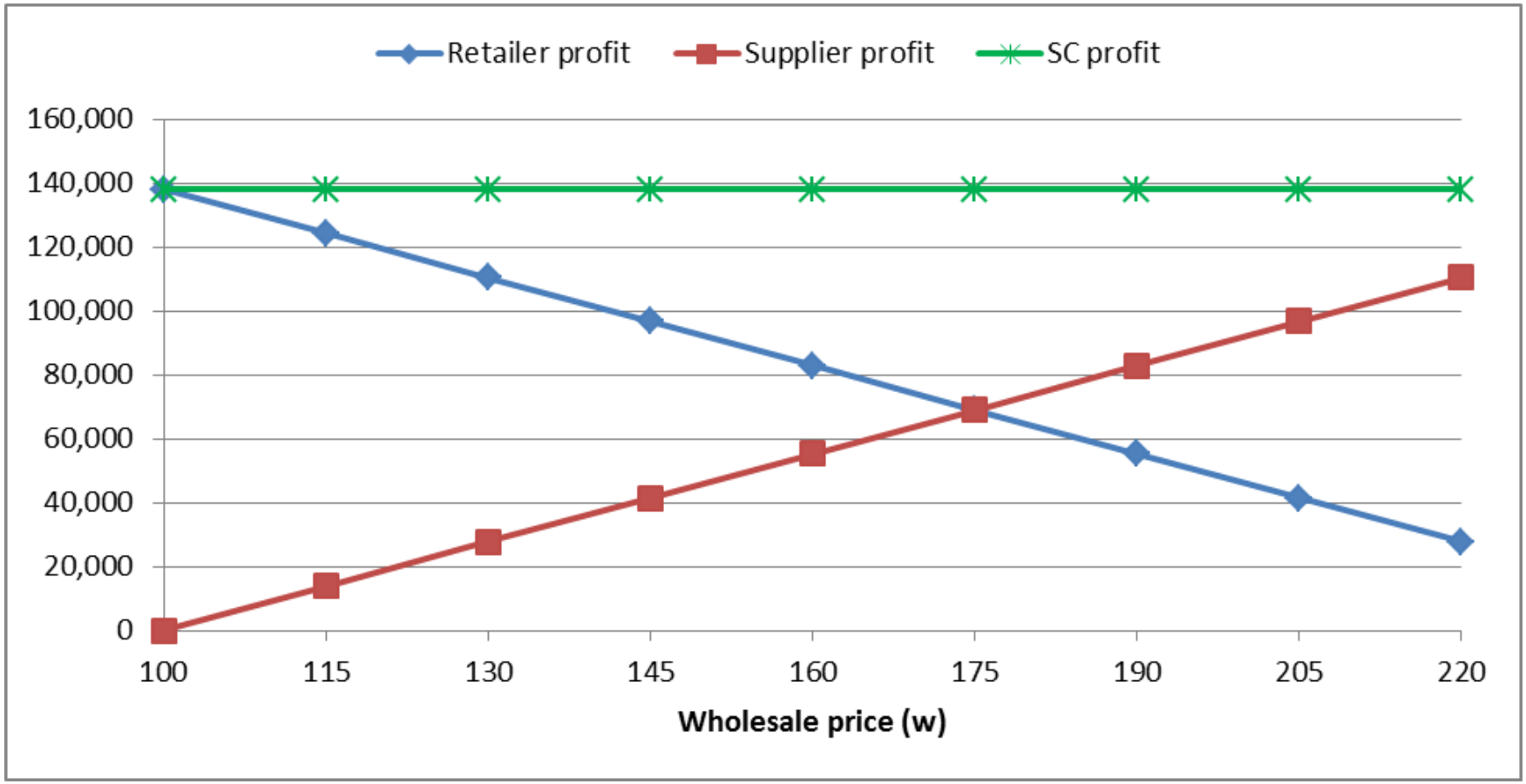
- Assume that Obermeyer sets w = $175 and wants to determine b.
- SKI PRO orders the optimal (profit maximizing) quantity:
  - $C_u = p - w = 250 - 175 = 75$
  - $C_o = w - b + r = 180 - b$
- Obermeyer chooses b to make SKI PRO's critical ratio equal the integrated supply chain's critical ratio:

$$\frac{p-w}{p-b+r} = \frac{p-c}{p-s} \Rightarrow b = p + r - (p-w)\frac{p-s}{p-c} = \$167.5$$

- With w=$175 and b=$167.5, SKI PRO is induced to order the optimal quantity (profit maximizing for the integrated supply chain).
  - However, for this contract to be feasible, both parties must be better off (with regard to alternative contracts).

Ski Jacket: Potential allocations of profit

Buy-back price (not shown here) is set to optimal (is derived from the formula)



Supply Chain profit w/o buyback: $125,342

Obermeyer profit w/o buyback: $70,950

SKI PRO profit w/o buyback: $54,392

Buy-Back contracts: Application and limitations

- How do they improve supply chain performance?
  - ✓ Allows for the redistribution of inventory risk across the supply chain.
- Requires suppliers to have an effective reverse logistics systems.
- Costs involved: Administrative and material handling.
- Provides retailers with incentives to push competing products from suppliers with whom they do not have buy back agreements.
- Accounting: Should products sold under a buy-back contract be considered as sales?
- Industries applied: audio, books, press.

Other solutions to supply chain coordination

- **Quantity discounts**:
  - Used to induce larger downstream order quantities so that downstream service is improved and/or handling and transportation efficiency is improved.
- **Revenue sharing contracts**:
  - If the manufacturer reduces wholesale price, the retailer can share a percentage of the revenue.
- **Quantity-flexibility contracts**:
  - The supplier provides full refund for returned (unsold) items as long as the number of returns is no larger than a certain quantity.

Revenue Management with Capacity Controls

Matching supply to demand when supply is fixed

- Examples of fixed supply:

  Travel industries (fixed number of seats, rooms, cars, etc.).

  Advertising time (limited number of time slots).

  Telecommunications bandwidth.

  Size of an MBA program.

- Revenue management is a solution if:
  - Adjusting supply is impossible (adjust the demand!)
  - Customers can be segmented according to their willingness to pay for the goods or services.
  - Number of items sold at a low price can be controlled (i.e., average price will depend on the mix of customers).

Environments suitable for revenue management

- The same unit of capacity (e.g., airline seat) can be used to deliver services to different customer segments (e.g., business and leisure customers) at different prices.



- Capacity is perishable (cannot be stored) and limited (all possible customers cannot always be served).
- Capacity is sold in advance of demand.
- There is an opportunity to segment customers (so that different prices can be charged) and different segments are willing to pay different prices.

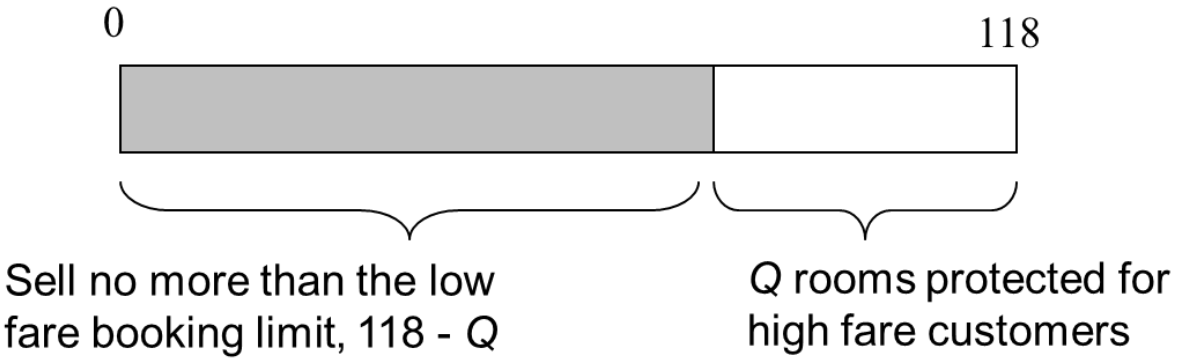
Practical problem



- Consider the Park Hyatt Philadelphia at the Bellevue, which has a capacity of 118 King/Queen rooms.
  - Hyatt offers a discount fare for a mid-week stay targeting leisure travellers.
  - There is also a regular fare (high fare) targeting business travellers.
- Demand for low fare rooms is abundant and it usually arrives early.
- **Let D be the uncertain demand for high fare rooms.**
  - Assume most of the high fare (business) demand occurs only within a few days/weeks of the actual stay.
- Objective:
  - **Maximize expected revenues by controlling the number of low fare rooms you sell.**

Yield management decisions

- The **booking limit** is the number of rooms you are willing to sell in a fare class or lower.
- The **protection level** is the number of rooms you reserve for a fare class or higher.
- Let **Q be the protection level** for the high fare class.
- Since there are only two fare classes, the booking limit on the low fare class is 118 – Q:
- You will sell no more than 118-Q low fare rooms because you are protecting (or reserving) Q rooms for high fare passengers.



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The connection to the newsvendor

- A single decision is made before uncertain demand is realized.
- There is an overage cost:
  - If D < Q then you protected too many rooms (you over protected)... so some rooms are empty which could have been sold to low fare travellers.
- There is an underage cost:
  - If D > Q then you protected too few rooms (you under protected)… so some rooms could have been sold at the high fare instead of the low fare.
- Choose Q to balance the overage and underage costs.

Revenue Management:

Overbooking

Ugly reality: cancellations and no-shows

- Depending on the industry, a relatively large number of reservations get cancelled at some point in time.
- In many cases (car rentals, hotels, full fare airline passengers) there is no penalty for cancellations (which almost invites “no-shows”).
- Problem:
  - The company may fail to fill its capacity (e.g., seat, room, etc.) if a passenger cancels at the very last minute or does not show up.
- Solution:
  - Accept/sell more reservations than capacity.
- Danger:
  - Some customers may have to be denied service even though they have a confirmed reservation.

Hyatt's Problem – Connection to the newsvendor

- The forecast for the **number of customers that do not show up** (let X) follows a known distribution.
- There is a cost of denying a room to the customer with a confirmed reservation (goodwill and penalties).
- Question: **How many rooms (let R) should be overbooked (sold in excess of capacity)?**
- Newsvendor setup:
  - Single decision where the number of no-shows is uncertain.
  - Underage cost if X > R (insufficient number of rooms overbooked).
  - Overage cost if X < R (too many rooms overbooked).

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Readings

- Chapter 16 and 17 (sections 17.3 and 17.4) from Cachon & Terwiesch, 3rd edition

Thank you!